CLASS 11-CBSE MATHEMATICS

CHAPTER: COMPLEX NUMBERS & QUADRATIC EQUATIONS-SET1

TIME : 1Hr **Mx.Marks:30**

SECTION A

Q.I Answer the following questions **[1X3 = 3 MARKS]**

1. Find The reciprocal of $3+\sqrt{7}i$.
2. Find the multiplicative inverse of the complex number $4 - 3i$
3. Solve the equation $x^2 + 3 = 0$

SECTION B

Q.II Answer the following questions **[2X5 = 10 MARKS]**

4. Express the given complex numbers in the form $a + ib$:
   a)$5i(-\frac{3}{5}i)$ b)$i9 \ + \ i19$
5. Convert the given complex number in polar form: $\sqrt{3} + i$
6. The value of $(1+i)^6+(1-i)^6$ is
7. Find the principal argument of $\left(1 + i\sqrt{3}\right)^2$.
8. Find the solution of the equation $x+\frac{1}{x}=2$.

SECTION C

Q.III Answer the following questions **[3X3 = 9 MARKS]**

9. Find the argument of $\frac{1-i}{1+i}$
10. If $x - iy = \sqrt{\frac{a-ib}{c-id}}$ then prove that $\left(x^2 + y^2\right)^2 = \frac{a^2+b^2}{c^2+d^2}$

11. Solve $3x^2 - 4x + \frac{20}{3} = 0$

SECTION D

Q.IV Answer the following questions **[4X2 = 8 MARKS]**

12. Solving the following quadratic equations by factorization method:

(i) $x^2 + 10ix - 21 = 0$

(ii) $x^2 + (1 - 2i)x - 2i = 0$

(iii) $x^2 - \left(2\sqrt{3} + 3i\right)x + 6\sqrt{3}i = 0$

(iv) $6x^2 - 17ix - 12 = 0$

13. If $f(z) = \frac{7-z}{1-z^2}$, where $z = 1 + 2i$, then find $|f|z||$.